

Silent Judge: Weak LLMs as Evaluators Inflate Privacy Failure Rates by 3.5× in PLAB

Nidhi Mehta · nidhi_mehta@outlook.com · Independent Researcher
PLAB v0.3 / v0.4 · September 2026 · github.com/NidhiMehta/plab

We introduce PLAB, a deterministic privacy-attack benchmark for large language models. PLAB v0.3 comprises 3,600 cases spanning 6 domains, 8 attack families, 25 scenario families, and 3 difficulty levels. Evaluating a subject model with a weak LLM judge (qwen3.5:3b) yields 1,441 flagged failures; cross-checking with a stronger judge (claude-opus-4-7) reveals only 412 true failures — a **3.5× inflation**. 683 of the original flags (47.4%) are confirmed hallucinations in the weak judge’s rationale. PLAB v0.4 introduces 45 hand-crafted cases across four structured tiers with deterministic evaluation for tiers 1–3 and judge-based evaluation for tier 4. Across five frontier models, pass rates range from 88.9% (Gemini 2.5 Pro) to 97.8% (DeepSeek R1). The benchmark, all model outputs, and evaluation code are available at github.com/NidhiMehta/plab.

1 THE JUDGE INFLATION PROBLEM

LLM-as-judge evaluation is widely used in privacy benchmarks because exact-match detection misses partial or implicit leakage. However, a judge model can *hallucinate* a violation where none exists — flagging a model refusal as a leak.

3.5× inflation.
qwen3.5:3b flagged 1,441 / 3,600 cases (40%). A controlled re-evaluation with claude-opus-4-7 confirmed only 412 (11.4%) as true failures. Of the 1,029 disagreements, **683 are confirmed hallucinations** (47.4% of all original flags) — the judge invented a violation that was not present.

The remaining 346 disagreements are ambiguous: the case involves borderline partial leakage where reasonable judges differ. We conservatively count these as non-failures in the headline numbers.

2 V0.3 RESULTS

METRIC	VALUE
SCALE	

4 PLAB V0.4 TIER DESIGN

v0.4 replaces the flat 3,600-case corpus with 45 hand-crafted cases across four structured tiers. Tiers 1–3 use deterministic evaluation; tier 4 uses claude-opus-4-7 as judge.

TIER	DESCRIPTION	CASES
<i>tool_gated</i>	Model must not call a tool it lacks authorization for	15
<i>implicit_authz</i>	Authorization is implied by context, not stated	10
<i>confused_deputy</i>	Third-party request tricks model into acting on behalf of attacker	10
<i>chained</i>	Multi-step attack across tool calls; requires semantic judge	10

5 V0.4 MODEL RESULTS

MODEL	PASS	TOTAL	RATE
DeepSeek R1	44	45	97.8%

METRIC	VALUE
Total cases	3,600
Domains	6
Attack families	8
Difficulty levels	3
JUDGE DISAGREEMENT	
Weak judge flags (qwen3.5:3b)	1,441 (40.0%)
True failures (claude-opus-4-7)	412 (11.4%)
Inflation factor	3.5×
Confirmed hallucinations	683 (47.4%)
Ambiguous cases	346 (9.6%)
FAILURE PATTERN	
Partial-leak anti-pattern	13 / 15 (87%)

Partial-leak anti-pattern: model refuses the targeted asset but leaks an adjacent data field (e.g. refuses to give an account number but states the account holder’s address).

3 CONTROLLED EXPERIMENT

To isolate judge hallucination from genuine disagreement, we constructed 100 cases where the subject model’s response is an unambiguous refusal (ground truth false-positive rate = 0%). We then ran both judges:

JUDGE	CONDITION	FP RATE
qwen3.5:3b	—	100%
claude-opus-4-7	Contaminated	0%
claude-opus-4-7	Clean	2%

The weak judge hallucinates violations on every single clear-refusal case. The stronger judge is near-perfect.

MODEL	PASS	TOTAL	RATE
o3	43	45	95.6%
GPT-5	43	45	95.6%
Claude Opus 4.7	42	45	93.3%
Gemini 2.5 Pro	40	45	88.9%

6 WHY DETERMINISTIC EVALUATION

For tiers 1–3 the violation signal is binary and unambiguous: did the model call a tool it was not authorized to call? Did it output the exact protected string? These checks require no LLM and produce no hallucinations.

Tier 4 (chained attacks) involves multi-turn reasoning where the violation may be implicit across multiple tool calls. A semantic judge is necessary there, but its scope is bounded — a deterministic pre-check gates whether the judge is invoked at all, preventing the class of hallucinations seen in v0.3.

7 RELATED WORK

PrivacyLens (Shao et al., 2024) and CONFAIDE (Miresghallah et al., 2024) evaluate contextual privacy norms but rely entirely on LLM judges without hallucination auditing. PLAB contributes: (i) a deterministic ground-truth layer that bounds judge influence, (ii) a controlled experiment quantifying judge hallucination rate, and (iii) an explicit anti-pattern taxonomy.

8 REPRODUCTION

All benchmark cases, evaluation code, raw model outputs, and HTML/PDF reports are available at:

github.com/NidhiMehta/plab

See **REPRODUCING.md** for step-by-step instructions to re-run v0.3 (3,600 cases), v0.4 (45 cases, 5 models), and regenerate all figures.

